

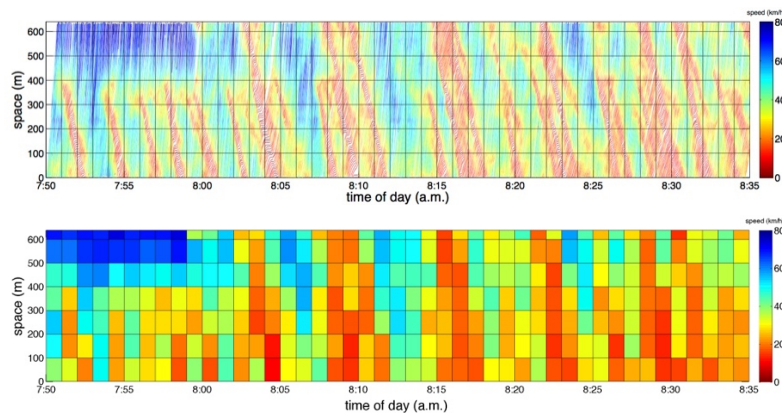
Constructing spatiotemporal speed contour diagrams: using rectangular or non-rectangular parallelogram cells?

Zhengbing He

webpage: www.GoTrafficGo.com . email: he.zb@hotmail.com

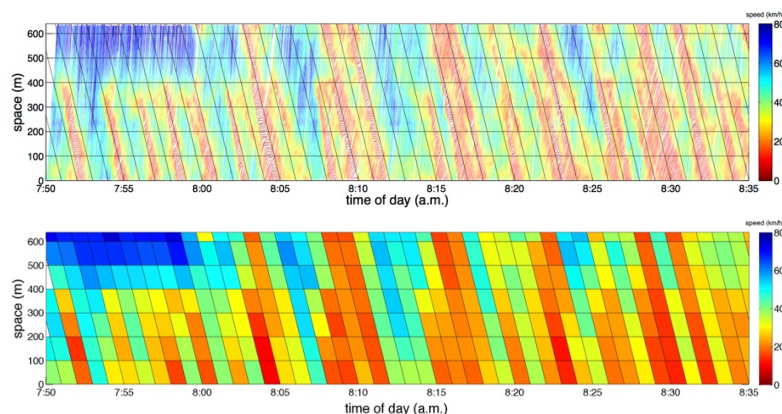
Gaps in Existing Studies

Time-space diagram of traffic is a very basic but important tool of understanding traffic. Traditional way of constructing the diagram is based on **rectangular time-space cells**, i.e., aggregating traffic speed in rectangular time-space cells.



Contribution

This paper empirically shows that using **non-rectangular parallelogram cells**, which considers **traffic waves**, to construct time-space diagram is **BETTER**.



Shining Points

The idea of considering traffic waves is straightforward. The time-space diagram of traffic is basic. However, nobody ever puts them together. These facts, in addition to the importance of the diagram, make the study unique and potentially very useful.



Constructing spatiotemporal speed contour diagrams: using rectangular or non-rectangular parallelogram cells?

Zhengbing He^a, Ying Lv^a, Lili Lu^{b,c} and Wei Guan^a

^aSchool of Traffic and Transportation, Beijing Jiaotong University, Beijing, People's Republic of China; ^bFaculty of Maritime and Transportation, Ningbo University, Ningbo, People's Republic of China; ^cJiangsu Province Collaborative Innovation Center for Modern Urban Traffic Technologies, Nanjing, People's Republic of China

ABSTRACT

A spatiotemporal speed contour (SSC, or time-space traffic) diagram that exhibits traffic dynamics in time and space is of importance in transportation research and applications. This paper empirically investigates the feasibility of using non-rectangular parallelogram (nRP) cells to construct the SSC diagram, given complete trajectory and probe vehicle data. Compared with the traditional way of using rectangular parallelogram (RP) cells, using nRP cells further considers the direction of backward-moving wave. The trajectory dataset provided by Next Generation Simulation is employed as the data source, and travel time is taken as a metric to measure the quality of the constructed SSC diagram. Various comparisons demonstrate that using nRP cells outperforms using RP cells, in particular when the traffic is congested and the cells are relatively large. Therefore, we recommend using nRP cells to construct the SSC diagram in future, which is as simple as the traditional way but more accurate particularly in terms of estimating travel time.

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Spatiotemporal speed contour diagram; time-space traffic diagram; traffic wave; travel time; probe vehicle data

1. Introduction

A spatiotemporal speed contour (SSC or time-space traffic) diagram, in which x-axis is time, y-axis is space, and the color inside (or z-axis) represents speed, is a foundation of various transportation research and applications, such as identifying bottlenecks (Chen, Skabardonis, and Varaiya 2004; Ban, Chu, and Benouar 2008), and understanding traffic characteristics (Treiber, Kesting, and Helbing 2010; He, He, and Guan 2015). Usually, the SSC diagram is constructed by virtually partitioning a spatiotemporal plane into homogeneous cells (or regions), and then estimating traffic speed or conditions of all the cells through various inference methods.

The most prevailing way is still to horizontally and vertically partition the spatiotemporal plane, that is, based on homogeneous rectangular parallelogram (RP) cells; see, for example, Nanthawichit, Nakatsuji, and Suzuki (2003), Chen, Skabardonis, and Varaiya (2004), Ban, Chu, and Benouar (2008), Wilson (2008), Wiecek, Fernández-Moctezuma, and Bertini (2009), Work et al. (2010), Zheng et al. (2011), Bucknell and Herrera (2014), Patire et al. (2014), Piccoli et al. (2015), He et al. (2017), and the research of the well-known Mobile Century experiment (Herrera et al. 2010; Herrera and Bayen 2010). Moreover, travel times are usually directly estimated based on the constructed SSC diagram (Chen, Rakha and Sadek 2011; Chen and Rakha 2012; Yildirimoglu and Geroliminis 2013; Chilukuri, Laval, and Guin 2014; Elhenawy, Chen, and Rakha 2014; Zhang, Wang and Chen 2017). However, the naive method of constructing the SSC diagram usually results in higher wave speed, that is, a vertical trend

instead of being along wave direction emerges in the reconstructed traffic waves (one may refer to Figure 2(c) in Bucknell and Herrera 2014 as an example).

Recent research has demonstrated the advantage and importance of analyzing and modeling traffic characteristics along the direction of backward-moving wave, whose speed is well known to be constantly $-10 \sim -20$ km/h around the world (Windover and Cassidy 2001; Mauch and Cassidy 2002; Ahn and Cassidy 2007; Schönhof and Helbing 2007; Laval and Leclercq 2010; Zheng et al. 2011; Chen et al. 2014). Aggregating traffic data along the wave direction is easier to meet the requirement of homogeneous traffic conditions of Edie's generalized definition (Edie 1965), where macroscopic traffic variable, that is, speed, flow and density, are computed based on an area in the SSC diagram instead of time (aerial photograph) or space (loop detector) cross-section. New insights on the basic traffic flow theory are gained along the wave direction. For example, detected vehicle speed was shifted along the wave direction, and travel time could be accurately estimated only based on the data collected by dual loop detectors (Coifman 2002). A new hysteresis pattern different from the pattern resulted from the traditional horizontal and vertical way was obtained (Laval 2011). By considering the wave direction, a new method to construct the fundamental diagram was also proposed (Chiabaut, Buisson, and Leclercq 2009). Moreover, Treiber and Helbing (2002), Treiber, Kesting, and Wilson (2011), and Van Lint and Hoogendoorn (2010) proposed and extended an adaptive filter to interpolate traffic conditions, and the results showed that the filter considering forward and backward wave speed outperformed the conventional method without such consideration.

In summary of the existing transportation research, the importance of analyzing traffic along the wave direction has been demonstrated, while a great amount of research still horizontally and vertically partitions the spatiotemporal plane to construct the SSC diagram. To improve the basic but prevailing method, this paper investigates the feasibility of partitioning the spatiotemporal plane along the wave direction to construct the SSC diagram, that is, using non-rectangular parallelogram (nRP) cells, whose left and right edges are along the wave direction. Using the nRP cells is still as simple as using the RP cells, and it further considers the speed of the wave, which is more consistent with real traffic characteristics.

Considering practical importance of extracting travel time from the SSC diagram, this paper first estimates travel times from the constructed SSC diagram, and then evaluates the quality of the construction by using those travel times.

Due to the popularity of probe (vehicle) data in practice (which are widely used by many researchers and map-based service providers, such as Google Map), the paper compares the effectiveness of using the RP and nRP cells under the condition that complete trajectory and partial probe data are available. Constructing the SSC diagram by using RP cells is a typical straightforward method of reconstructing traffic state. Although we have already had advanced methods to estimate travel time based on low frequency probe vehicle data, such as those proposed in Jenelius and Koutsopoulos (2013), extracting travel time through constructing the SSC diagram is also a simple but important alternative. Thus, it is significant to study if simply changing the RP cells to the nRP cells could improve the quality of the construction of the SSC diagram.

The remainder of the paper is organized as follows: Section 2 describes the data, and the methods of constructing the SSC diagram and estimating travel time; Sections 3 and 4 evaluate the effectiveness of using RP and nRP cells, when complete trajectory and partial probe data are provided, respectively; Section 5 compares the results obtained in Sections 3 and 4; Section 6 analyzes the sensitivity of using nRP cells with different wave speeds; Conclusions are made at last.

2. Data and methodology

2.1. Data description

This paper adopts one of high-fidelity trajectory datasets provided by Next Generation Simulation program (NGSIM), which was collected on a 6-lane segment in the vicinity of Lankershim Avenue on

southbound US-101 freeway in Los Angeles, California. The time period of data collection ranges from 7:35 am to 8:35 am on 15 June 2005. We select the trajectory dataset collected on Lane 1 (median lane), 2, 3, and 4, where stop-and-go oscillations originate and propagate upstream at a constant wave speed of approximately -16 km/h (Laval and Leclercq 2010; He, Zheng, and Guan 2015). The original dataset provides a data sample every 0.1 s. This paper simply averages these data samples every 1 s (arithmetic mean) to smooth out the detection errors (for the feature of the errors, see e.g. Thiemann, Treiber, and Kesting 2008; Punzo, Borzacchiello, and Ciuffo 2011).

It is known that the traffic in the US-101 dataset was in congestion with stop-and-go oscillations, except a few faster vehicles that appeared at the beginning of the collection and were leaving the road section. Since the dataset exhibits various traffic patterns, empirical study based on the dataset might be representative. Although free-flow conditions are relatively lacking, congested conditions are definitely more important in practice.

The exact traffic conditions provided by the trajectories allow us to concentrate on the construction of the SSC diagram. In addition, through sampling the trajectory data, we can derive analog probe data with various penetration rates and sampling frequencies. It is convenient to evaluate the method supposing only probe data are available.

To be convenient, the first 600 m out of the total 640 m road section provided is selected, and it is known that the collection lasted for 45 min. Thus, the spatiotemporal plane is 600 m in height and 2700 s in width. About 5787 vehicles (automobiles) were traversing on Lane 1, 2, 3, and 4 during the study 45 min. We select the vehicles whose trajectories are longer than 590 m between the road section of 0 and 600 m, and take the travel time of these vehicles as the ground truth travel time. The total number of such vehicles is 2035, and their travel time ranges from 35 to 115 s, which turns out the average speed of about 61.7–18.8 km/h.

2.2. Penetration rate and sampling frequency of probe data

In practice, high-fidelity vehicle trajectories are precious, while probe vehicles collecting partial traffic information widely exist, such as t -axis and buses that are able to report their positions every short time interval. Thus, probe data are one of the most important and cheapest data sources to monitor the traffic.

To measure the density of probe data, we define a penetration rate (denoted by $p \in [0, 1]$) as the percentage of probe vehicles in total vehicles, and a sampling interval (denoted by s) as the time interval that probe vehicles upload their status information, usually including instantaneous speed, GPS locations, etc. Thus, sampling frequency can be defined as $z = 1/s \in [0, 1]$, that is, uploading times per sec. In practice, the penetration rate is usually lower than 10%, and the sampling interval may be ranging from 20 s to 5 min.

The density of probe data (denoted by k) heavily depends on p and z , and thus we propose to qualify the density by using

$$k = p \times z \in [0, 1], \quad (1)$$

which is similar to the metric given in Bucknell and Herrera (2014). It is straightforward that the greater the value of k is, the denser the probe data are; it is complete trajectory data for all vehicles, when $k = 1$.

To imitate the probe data with p and z , we randomly select a vehicle with a probability of p from the US-101 dataset as a probe vehicle. For the selected probe vehicle we randomly select a time point between 1 and s as the first time of sampling, and then the probe vehicle reports its position and speed every s sec, that is, every s sec the position and speed on the trajectory are taken into calculation.

2.3. Constructing SSC diagrams

According to Edie's generalized definition (Edie 1965), only when homogenous traffic states are contained by cells, the resulting average traffic variables are stationary and able to reflect the fundamental

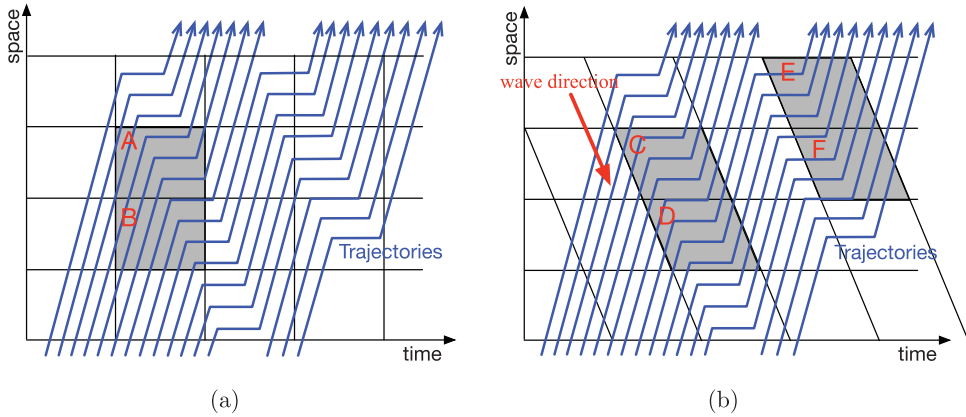


Figure 1. Schematic diagrams of vehicle trajectories for constructing the SSC diagrams: (a) using RP cells; (b) using nRP cells.

(equilibrium) relations among speed, flow, and density. Therefore, when constructing the SSC diagram, it is of importance to make the traffic states contained by each cell as homogenous as possible.

A native method of constructing the SSC diagram is to virtually partition a spatiotemporal plane into homogenous cells, and then to take the average of the speed of all vehicles contained by a cell as the speed of the cell. Traditionally, homogeneous RP cells are used; see Figure 1(a) for an illustration. This paper evaluates the feasibility of using nRP cells, the slopes of whose left and right edges are equal to the wave speed; see Figure 1(b).

Comparing Figure 1(b) with Figure 1(a), it can be seen that in this case at similar locations nRP cells C and D in Figure 1(b) contain similar complete processes of deceleration and acceleration traffic, while RP cells A and B in Figure 1(a) do not. Generally speaking, nRP cells have more chances to contain complete stop-and-go processes as well as homogenous traffic. Although a stop-and-go process may not be completely captured by an nRP cell limited by its width (see cells E and F in Figure 1(b)), the traffic contained by the spatial-neighbor nRP cells are more likely similar. It means that the nRP cells are beneficial to capturing similar traffic waves. Thus, we intuitively conjecture that using nRP cells outperforms using RP cells in constructing the SSC diagram, as well as further estimating travel time.

In the paper, we construct the SSC diagram by using the cells with various widths (denoted by w) and heights (denoted by h). Considering the fact that the penetration rate of probe data is usually lower than 10% and the sampling interval may be ranging from 20 s to 5 min, we then select that $w = \{10, 30, 60, 90\}$ s and $h = \{10, 50, 100, 150\}$ m, which will not result in a large scale of empty cells appearing in the SSC diagram. Moreover, larger cells are not meaningful, because they may erase significant traffic dynamics.

2.4. Estimating travel time via the SSC diagram

To extract travel time from the SSC diagram, we first reconstruct the trajectory of a vehicle, by setting the slope of the trajectory section that is within a cell be equal to the estimated speed of the corresponding cell. Then, the time difference between the starting and ending times of the trajectory is taken as the estimated travel time. See Figure 2 for a schematic diagram of the method.

The method is simple but widely used, for example, Chen, Rakha and Sadek (2011), Chen and Rakha (2012), Yildirimoglu and Geroliminis (2013), Elhenawy, Chen, and Rakha (2014), Chilukuri, Laval, and Guin (2014) and Zhang, Wang and Chen (2017). The purpose of estimating travel time here is to measure the quality of the constructed SSD diagram, and thus the simple method without strong artificial intervene might be more proper.

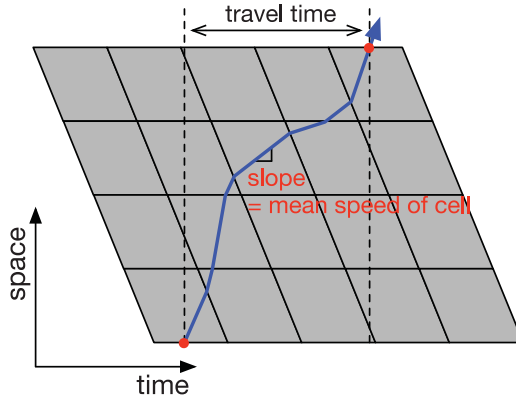


Figure 2. A schematic diagram of estimating travel time via the SSC diagram.

2.5. Filling empty cells in the SSC diagram

Since probe data usually sparsely exist, it is not guaranteed that every cell can be occupied by at least one data point, meaning that empty cells without mean speed may exist in the SSC diagram. In particular, when the penetration rate or the sampling frequency is low, or the width or height of the cell is small, the number of empty cells is not negligible. The empty cells hinder travel time estimation, and we have to fill them. Since filling the empty cells is not the focus of this paper, a simple method that does not introduce strong artificial intervene is adopted, that is, simply taking the average of the speed of all neighbors as the speed of the empty cell.

More specifically, there are eight neighbors for a cell that is not on the edge of the SSC diagram. It is easy to fill all empty cells whose neighbors are all filled with speed, by taking average of the speed of the neighbors. However, due to the possibility that empty neighbors exist, an iteration is needed in averaging empty cells with empty neighbors.

See Figure 3 for a typical case in which an empty cell (cells A and C in Figure 3(a)) has one empty neighbor (cell B in Figure 3(a)) while the empty neighbor has more empty neighbors (cells A and C in Figure 3(a)). To fill such empty cells, we first fill all empty cells with only one empty neighbor, by taking average of the speed of the other neighbors, that is, filling cells A and C in Figure 3(a) by averaging its seven neighbors having speed. After this round, the empty neighbors of a part of empty cells (cell B) become less (see Figure 3(b)). Then, we go back to fill all empty cells, starting from filling the empty cell without empty neighbors (cell B in Figure 3(b)). Repeat the process that filling empty cells with more empty neighbors (starting from the cell with one empty neighbor, and then those with two empty neighbors), and go back to fill all empty cells starting from those without empty neighbors, until all empty cells are filled.

This iterative process considers the correlation between the empty cell and its neighbors at the greatest extent. In spite of simplicity, the native method may fail when the number of empty cells is large, because there is no enough neighbor data to be averaged.

2.6. Qualifying comparison

To make quantitative comparison, we estimate travel time based on the constructed SSC diagram. To make it comparable, the travel time is estimated beginning from the starting time of a real trajectory, and the mean absolute percentage errors (MAPEs) for all estimated travel time are calculated to qualify the estimation. The formula of the MAPE can be written as follows

$$R_C = \frac{1}{n} \sum_{i=1}^n \frac{|\hat{\tau}_{C,i} - \tau_i|}{\tau_i} \cdot 100\%, \quad (2)$$

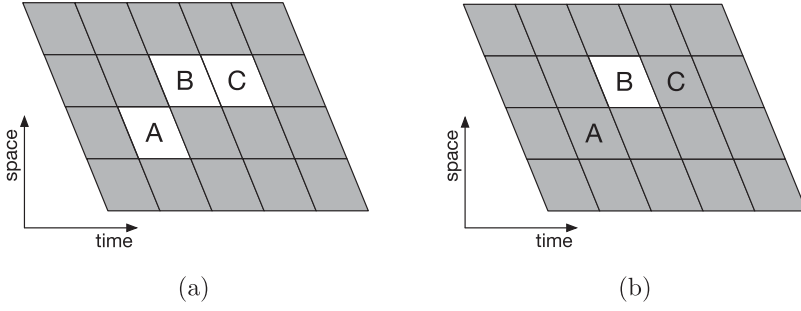


Figure 3. An example of filling empty cells by taking average iteratively. (a) The initial condition, where A and C have one empty neighbor (B), and B has two empty neighbors (A and C); (b) After filling all empty cells with only one empty neighbors (A and C), B turns to the empty cell without empty neighbors.

where $C = \{RP, nRP\}$ represents RP and nRP cells, respectively; $\hat{\tau}_{C,i}$ and τ_i are the estimated and ground truth travel times, which have the same starting time on the road section; n is the total number of the trajectories used for the comparison (as mentioned in Section 2.1, $n = 2035$ for the selected dataset). Meanwhile, we compute the difference of the MAPE between the estimations of using RP and nRP cells, that is,

$$\Delta R = R_{RP} - R_{nRP}. \quad (3)$$

If $\Delta R > 0$, we say that using nRP cells is better than using RP cells, and vice versa.

3. RP vs. nRP cells: comparison provided complete trajectory data

This section makes an accurate empirical comparison between the SSC diagrams constructed by using RP and nRP cells. To this end, we consider all US-101 trajectories, that is, suppose to know all traffic information, and average all speed of all vehicles contained by a cell, after respectively partitioning the US-101 spatiotemporal plane by using RP and nRP cells with various sizes.

It is known that the wave speed is about -16 km/h in the US-101 dataset, and thus the slopes of the left and right edges of the nRP cells are set to be -16 km/h.

Taking Lane 1 as an example, Figure 4 presents the ground truth US-101 trajectories and its SSC diagrams constructed by using RP and nRP cells. Visually, as expected, nRP cells are better at reflecting the trend of traffic propagation along the wave direction. For detailed comparison, see the marked cells about 8:00 am for example, where the SSC diagram constructed by using nRP cells looks closer to real traffic than those by using RP cells. Although the SSC diagram constructed by using nRP cells also blurs the stop-and-go wave (see the marked cells about 8:10 am, for example), it still looks better than the SSC diagram constructed by using RP cells.

Further, Table 1 shows the detailed results when we construct the SSC diagram using various cells, provided complete trajectory data, and the following observations can be made:

Observation 3.1: It can be generally seen from Table 1 that $R_{RP} \in [1.3\%, 11.4\%]$, $R_{nRP} \in [1.1\%, 8.5\%]$, and $\Delta R \in [-1.4\%, 7\%]$; the values are marked in bold in Table 1. It means that the travel time estimation errors by using the SSC diagram is not large, when providing complete trajectory data.

Observation 3.2: Most ΔR are larger than zero when the traffic is in various states, that is, most R_{nRP} are smaller. Note that the number of such cases is 77 out of total 90 cases, and all the opposite cases are marked with underlines. It means that in most cases, using nRP cells is better than using RP cells.

Observation 3.3: In Columns 4 and 5, where the intervals are $(70, 85]$, $(30.9, 25.4]$, and $(85, \infty)$, $(25.4, 0)$, respectively, all ΔR are positive except one case. The maximum ΔR is around 5%, compared with

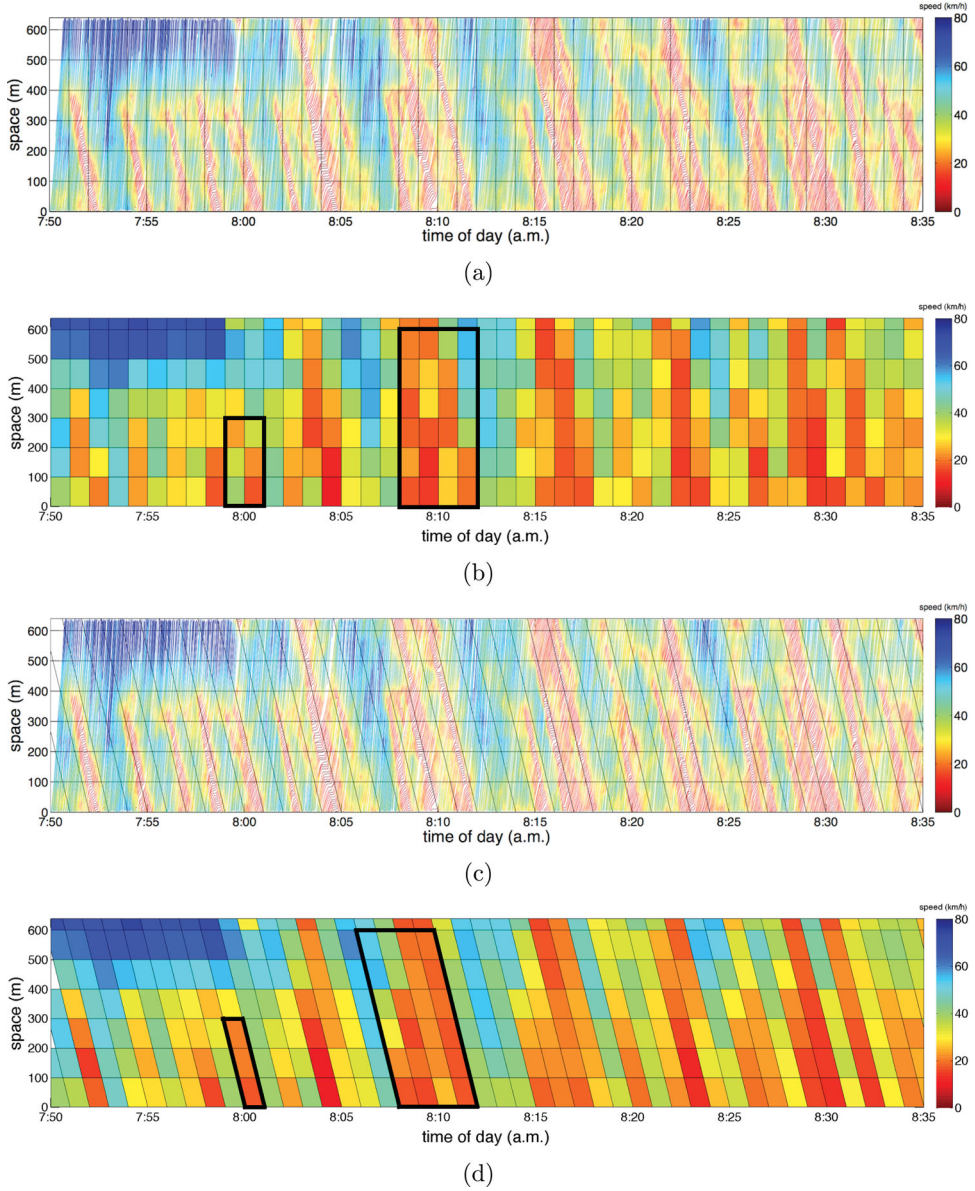


Figure 4. Illustrations of the constructed SSC diagrams, by taking Lane 1 on US-101 freeway in Los Angeles, as an example. The spatial and temporal resolutions are 100 m and 60 s, respectively: (a) The spatiotemporal diagram of trajectories partitioned using RP cells; (b) The resulting SSC diagram of subfigure (a); (c) The spatiotemporal diagram of trajectories partitioned using nRP cells; (d) The resulting SSC diagram of subfigure (c).

$R_{RP} \in (10\%, 11\%)$ and $R_{nRP} \in (4\%, 6\%)$, approximately. It means that when the traffic is in congestion, using nRP cells is much better than using RP cells. The advantage comes from the fact that the nRP cells better consider the congested waves.

Observation 3.4: In Columns 2 and 3 where the intervals are $[0, 55]$, $(\infty, 39.3]$ and $(55, 70]$, $(39.3, 30.9]$, respectively, more negative ΔR emerge, in particular when the cells are small. It means that when the traffic speed is relatively high and the cells are small, using nRP cells may be worse than using

Table 1. Comparison of MAPE (%) between the ground truth and the estimation (using RP and nRP cells) of travel time in different intervals of travel time and mean speed, provided complete trajectory data.

$w \times h$	R_{RP} , R_{nRP} , and ΔR in different intervals of travel time and mean speed ^a				
	(0, 55], (∞ , 39.3] ^b	(55, 70], (39.3, 30.9]	(70, 85], (30.9, 25.4]	(85, ∞), (25.4, 0)	All
$10 \times 10 = 100$	1.7, 1.6, 0.1	1.9, 1.8, 0.1	1.5, 1.4, 0.2	1.3, 1.1 , 0.2	1.6, 1.5, 0.1
$10 \times 50 = 500$	1.7, 2.1, -0.4	1.8, 2.2, -0.4	1.9, 1.7, 0.2	1.7, 1.3, 0.4	1.8, 1.9, -0.1
$10 \times 100 = 1000$	1.8, 2.8, -1.0	2.4, 3.0, -0.6	3.4, 2.3, 1.1	3.4, 2.1, 1.3	2.7, 2.6, 0.1
$10 \times 150 = 1500$	2.0, 3.4, -1.4	3.2, 3.5, -0.3	4.4, 3.0, 1.5	4.3, 2.4, 1.9	3.4, 3.1, 0.3
$30 \times 10 = 300$	3.7, 3.1, 0.6	4.5, 3.0, 1.5	4.6, 2.4, 2.3	3.9, 1.9, 1.9	4.2, 2.6, 1.5
$30 \times 50 = 1500$	3.6, 3.1, 0.5	4.3, 3.1, 1.2	4.3, 2.4, 1.9	3.6, 1.9, 1.7	4.0, 2.7, 1.3
$30 \times 100 = 3000$	3.5, 3.6, -0.1	4.0, 3.7, 0.3	3.8, 2.8, 1.0	3.1, 2.5, 0.7	3.6, 3.2, 0.4
$30 \times 150 = 4500$	3.4, 4.2, -0.8	3.6, 4.2, -0.6	3.6, 3.8, -0.2	3.2, 2.7, 0.5	3.4, 3.8, -0.3
$60 \times 10 = 600$	6.1, 5.3, 0.8	8.2, 5.5, 2.7	8.2, 3.7, 4.6	8.9, 3.4, 5.5	7.8, 4.5, 3.3
$60 \times 50 = 3000$	6.5, 6.0, 0.5	8.8, 6.7, 2.1	8.5, 4.4, 4.0	8.8, 3.4, 5.4	8.1, 5.2, 2.9
$60 \times 100 = 6000$	5.9, 5.5, 0.5	7.9, 5.5, 2.4	7.8, 3.6, 4.3	8.4, 3.5, 4.9	7.5, 4.6, 2.9
$60 \times 150 = 9000$	6.0, 6.1, -0.1	7.7, 6.3, 1.4	7.4, 4.1, 3.3	7.9, 3.2, 4.7	7.2, 5.0, 2.2
$90 \times 10 = 900$	8.8, 7.3, 1.5	10.5, 7.1, 3.4	10.8, 5.2, 5.6	11.2, 5.1, 6.2	10.3, 6.2, 4.0
$90 \times 50 = 4500$	9.8, 8.5 , 1.4	11.4, 8.5 , 2.9	10.9, 5.8, 5.1	10.9, 3.9, 7.0	10.7, 6.8, 4.0
$90 \times 100 = 9000$	9.2, 8.1, 1.1	10.7, 8.0, 2.8	10.7, 5.3, 5.4	11.0, 4.2, 6.8	10.4, 6.5, 3.9
$90 \times 150 = 13,500$	8.9, 8.2, 0.7	10.4, 7.9, 2.5	10.5, 5.3, 5.2	10.8, 4.3, 6.4	10.1, 6.5, 3.6

^aThe intervals are divided by the ground truth travel time.

^bThe former interval is the travel time interval (s), and the latter one is the mean speed interval (km/h).

RP cells. RP cells may be better at capturing forward-moving wave, whose speed is close to free-flow speed. However, when the cell is enlarged, using nRP cells is still better than using RP cells.

Observation 3.5: When the cells are relatively large such that the width is larger than 60 s (see the last eight rows), R_{RP} are quite larger than R_{nRP} , in particular when the speed is low. ΔR is even larger than 5% in the condition that R_{RP} is around 10%. It means that using nRP cells is much better than using RP cells when the cells are large.

Figure 5 visualizes an example using the cell of $w = 60$ sec and $h = 100$ m, which provides more details of the estimation and comparison. It is additionally noticed from the figure that more errors are larger than zero when the travel time is short, while more errors are smaller than zero when the travel time is long. It means that for using both shapes of cells travel time is more likely over-estimated when traffic speed is high, while travel time is more likely under-estimated when traffic speed is low.

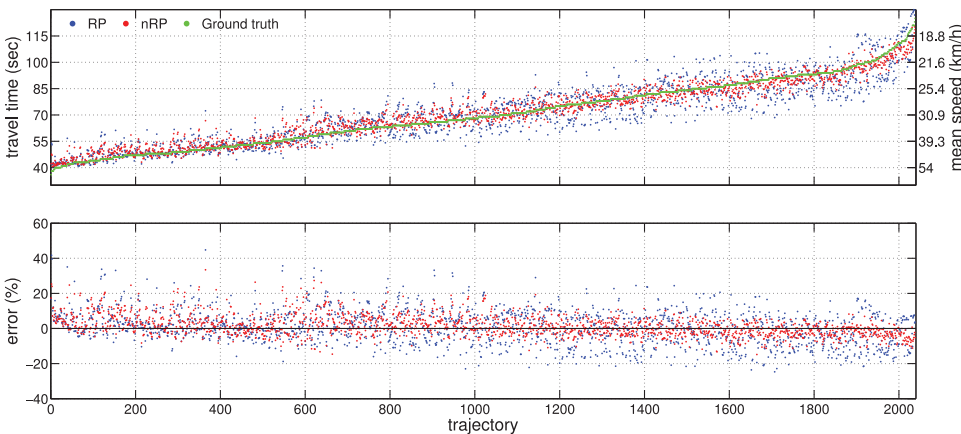


Figure 5. Comparison of travel time estimations based on all trajectory data, where the cells are 60 s in width and 100 m in height, respectively. The comparison is sorted by the ground truth travel time from small to large. Upper figure: Comparison of travel time; Lower figure: Comparison of percentage errors.

4. RP vs. nRP cells: comparison provided partial probe data

Given various penetration rates and sampling frequencies, we evaluate and compare the effectiveness of using RP and nRP cells. The paper considers the cases with $k = \{0.005, 0.010, 0.015, 0.020, 0.025, 0.03, 0.04, 0.05\}$, which are resulted from $p = \{0.1, 0.2, 0.3, 0.4, 0.5\}$ and $z = \{0.05, 0.10\}$, that is, $s = \{10, 20\}$ s. In each case, the RP and nRP cells with $w = \{30, 60, 90\}$ s and $h = \{50, 100, 150\}$ m are employed to construct the SSC diagram. The cells with $w = 10$ s or $h = 10$ m are discarded in the section, because such small cells will result in a large number of empty cells. The simple method of filling empty cells introduced in Section 2.5 may not work well, and the effectiveness may not be evaluated without a deviation. Twenty replications are made for each case, and their averages of R_{RP} , R_{nRP} , ΔR , which are denoted by $\bar{R}_{RP}$, $\bar{R}_{nRP}$, $\bar{\Delta R}$, are taken and analyzed. Table 2 presents the MAPE for the estimations. The observations can be made from the table as follows:

Observation 4.1: When the density of probe data is relatively small (see the cases when $k = 0.005$ in Table 2), more $\bar{\Delta R}$ (seven out of nine cases) is negative, indicating that using RP cells may be better than

Table 2. Comparison of MAPE (%) between the ground truth and the estimation (using RP and nRP cells) of travel time, provided partial probe data.

% of empty				% of empty			
$p \times z = k$	$w \times h$	cells	$\bar{R}_{RP}, \bar{R}_{nRP}, \Delta \bar{R}$	$p \times z = k$	$w \times h$	cells	$\bar{R}_{RP}, \bar{R}_{nRP}, \Delta \bar{R}$
$0.1 \times 0.05 = 0.005$	$30 \times 50 = 1500$	<u>72, 72</u> ^a	<u>11.7, 12.4, -0.7</u>	$0.5 \times 0.05 = 0.025$	$30 \times 50 = 1500$	<u>21, 22</u>	7.2, 7.2, 0.1
	$30 \times 100 = 3000$	<u>56, 56</u>	<u>10.2, 10.6, -0.4</u>		$30 \times 100 = 3000$	11, 11	6.0, 5.4, 0.6
	$30 \times 150 = 4500$	<u>47, 46</u>	<u>10.2, 10.4, -0.2</u>		$30 \times 150 = 4500$	10, 10	5.6, 5.0, 0.6
	$60 \times 50 = 3000$	<u>53, 53</u>	<u>16.1, 17.9, -1.8</u>		$60 \times 50 = 3000$	8, 9	10.3, 8.5, 1.8
	$60 \times 100 = 6000$	<u>34, 34</u>	<u>13.9, 14.7, -0.8</u>		$60 \times 100 = 6000$	6, 6	8.6, 6.2, 2.4
	$60 \times 150 = 9000$	<u>27, 26</u>	<u>12.5, 13.3, -0.8</u>		$60 \times 150 = 9000$	5, 6	7.9, 6.0, 1.9
	$90 \times 50 = 4500$	<u>40, 39</u>	<u>21.4, 21.6, -0.3</u>		$90 \times 50 = 4500$	5, 5	12.0, 8.3, 3.7
	$90 \times 100 = 9000$	<u>22, 22</u>	17.2, 15.5, 1.6		$90 \times 100 = 9000$	4, 4	10.8, 7.1, 3.7
	$90 \times 150 = 13,500$	18, 17	14.6, 12.7, 2.0		$90 \times 150 = 13,500$	4, 4	10.5, 7.1, 3.4
$0.2 \times 0.05 = 0.010$	$30 \times 50 = 1500$	<u>52, 53</u>	<u>9.7, 10.1, -0.4</u>	$0.3 \times 0.10 = 0.030$	$30 \times 50 = 1500$	17, 18	6.1, 6.1, 0.1
	$30 \times 100 = 3000$	<u>32, 33</u>	8.7, 8.4, 0.3		$30 \times 100 = 3000$	9, 9	5.1, 4.7, 0.3
	$30 \times 150 = 4500$	<u>25, 26</u>	8.4, 7.6, 0.8		$30 \times 150 = 4500$	9, 9	4.7, 4.5, 0.2
	$60 \times 50 = 3000$	<u>29, 30</u>	<u>14.6, 15.0, -0.4</u>		$60 \times 50 = 3000$	7, 7	9.2, 7.0, 2.2
	$60 \times 100 = 6000$	15, 15	11.7, 10.1, 1.6		$60 \times 100 = 6000$	5, 5	8.1, 5.6, 2.5
	$60 \times 150 = 9000$	12, 12	10.0, 8.4, 1.7		$60 \times 150 = 9000$	5, 5	7.5, 5.5, 2.0
	$90 \times 50 = 4500$	17, 18	18.3, 16.5, 1.8		$90 \times 50 = 4500$	4, 5	11.2, 7.4, 3.8
	$90 \times 100 = 9000$	9, 9	13.3, 10.4, 2.9		$90 \times 100 = 9000$	4, 4	10.5, 6.9, 3.6
	$90 \times 150 = 13,500$	8, 8	11.7, 8.8, 2.9		$90 \times 150 = 13,500$	4, 4	10.3, 6.9, 3.3
$0.3 \times 0.05 = 0.015$	$30 \times 50 = 1500$	<u>38, 38</u>	<u>8.4, 8.8, -0.4</u>	$0.4 \times 0.10 = 0.040$	$30 \times 50 = 1500$	11, 12	5.5, 5.2, 0.3
	$30 \times 100 = 3000$	<u>20, 21</u>	7.4, 7.1, 0.3		$30 \times 100 = 3000$	7, 7	4.6, 4.2, 0.4
	$30 \times 150 = 4500$	16, 17	7.0, 6.3, 0.7		$30 \times 150 = 4500$	7, 7	4.3, 4.2, 0.1
	$60 \times 50 = 3000$	17, 17	12.7, 12.2, 0.4		$60 \times 50 = 3000$	5, 5	8.5, 5.8, 2.7
	$60 \times 100 = 6000$	9, 9	9.6, 8.1, 1.4		$60 \times 100 = 6000$	5, 5	7.9, 5.1, 2.7
	$60 \times 150 = 9000$	8, 8	8.6, 7.1, 1.5		$60 \times 150 = 9000$	5, 4	7.4, 5.2, 2.2
	$90 \times 50 = 4500$	9, 9	14.5, 12.1, 2.5		$90 \times 50 = 4500$	4, 4	10.7, 6.8, 3.9
	$90 \times 100 = 9000$	5, 6	11.5, 8.1, 3.4		$90 \times 100 = 9000$	4, 4	10.4, 6.6, 3.8
	$90 \times 150 = 13,500$	5, 6	10.9, 7.7, 3.2		$90 \times 150 = 13,500$	3, 4	10.2, 6.7, 3.4
$0.4 \times 0.05 = 0.020$	$30 \times 50 = 1500$	<u>28, 29</u>	<u>7.8, 8.1, -0.2</u>	$0.5 \times 0.10 = 0.050$	$30 \times 50 = 1500$	8, 9	5.0, 4.7, 0.4
	$30 \times 100 = 3000$	14, 15	6.7, 6.0, 0.7		$30 \times 100 = 3000$	6, 6	4.3, 3.9, 0.4
	$30 \times 150 = 4500$	12, 13	6.1, 5.5, 0.6		$30 \times 150 = 4500$	6, 6	4.1, 4.0, 0.1
	$60 \times 50 = 3000$	11, 12	11.4, 10.1, 1.3		$60 \times 50 = 3000$	5, 5	8.3, 5.4, 2.9
	$60 \times 100 = 6000$	6, 7	9.0, 6.8, 2.2		$60 \times 100 = 6000$	4, 4	7.7, 5.0, 2.8
	$60 \times 150 = 9000$	6, 6	8.2, 6.4, 1.8		$60 \times 150 = 9000$	4, 4	7.3, 5.1, 2.2
	$90 \times 50 = 4500$	6, 6	12.9, 9.8, 3.1		$90 \times 50 = 4500$	4, 4	10.6, 6.5, 4.0
	$90 \times 100 = 9000$	5, 5	11.0, 7.5, 3.5		$90 \times 100 = 9000$	4, 4	10.3, 6.5, 3.8
	$90 \times 150 = 13,500$	4, 4	10.6, 7.3, 3.3		$90 \times 150 = 13,500$	3, 4	10.1, 6.6, 3.5

^aThe former and latter values are the percentages of empty RP and nRP cells, respectively.

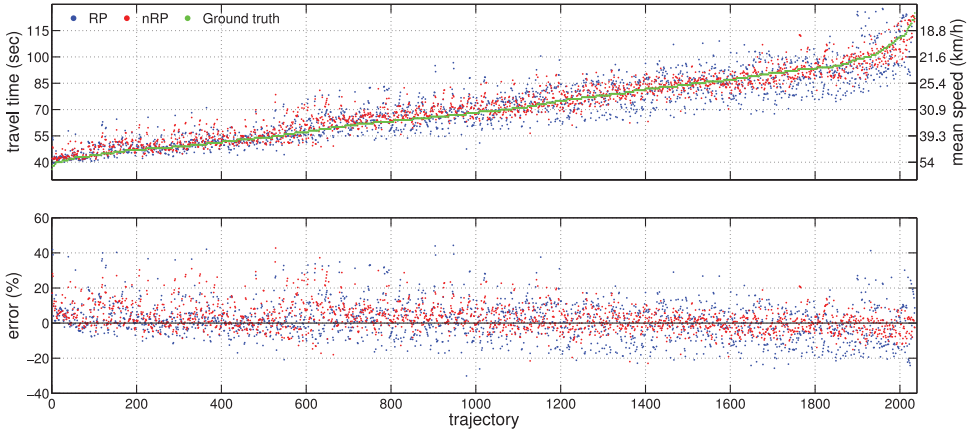


Figure 6. Comparison of travel time estimations based on probe data, where the spatial and temporal resolutions are 100 m and 60 s, respectively, and the penetration rate and sampling interval are 0.3 and 10 s, respectively. Upper figure: Comparison of travel time; Lower figure: Comparison of percentage errors.

using nRP cells when the probe data are sparse in density. However, large percentages of empty cells (ranging from 22% to 72%, which are marked with wavy lines) are noticed, meaning that the results may be biased by the simple method of filling empty cells introduced in Section 2.5. Thus, we cannot make a conclusion.

Observation 4.2: It can be generally seen from Table 2, except the cases whose percentage of empty cells is larger than 20%, $\bar{R}_{RP} \in [4.1\%, 18.3\%]$, $\bar{R}_{nRP} \in [3.9\%, 16.5\%]$, and $\Delta\bar{R} \in [0.1\%, 4\%]$; the values are marked in bold in Table 2. It means that the travel time estimation errors by using the SSC diagram is not large, when providing probe data.

Observation 4.3: Except the rows with large percentage of empty cells (larger than 20%), $\bar{R}_{RP}$ are greater than $\bar{R}_{nRP}$ on all rows. It clearly indicates the advantage of using nRP cells with various sizes in constructing the SSC diagram and estimating travel time.

Observation 4.4: When the cells are relatively large such that the width is larger than 60 s, the advantages are more obvious, where $\Delta\bar{R}$ is even as large as 2–4% in the condition that the total percentages are about 10%.

Similarly, Figure 6 visualizes an example using the cell of $w = 60$ s and $h = 100$ m, and $p = 0.3$, $z = 0.1$. The trend similar to Figure 5 is noticed, that is, more errors are larger than zero when the travel time is short, while more errors are smaller than zero when the travel time is long. It confirms that provided probe data, travel time is more likely over-estimated when traffic speed is high for using both shapes of cells, while travel time is more likely under-estimated when traffic speed is low.

5. RP vs. nRP cells: comparison provided complete and partial data

This section compares the results using complete trajectories and partial probe data; see Table 3. In the table, all cases have empty cells whose number is lower than 20%, that is, the rows marked with the wavy line in Table 2 are removed. We additionally calculate the standard deviation of all replications for each case to show the influence coming from randomly sampling, which are denoted by $\bar{\sigma}_{RP}$ and $\bar{\sigma}_{nRP}$, respectively. Note that the rows with $k = 1$ are the results using complete trajectories. It is observed from Table 3 as follows:

Table 3. Comparison of MAPE (%) between the estimations (using RP and nRP cells) of travel time, provided complete trajectory and partial probe data.

$w \times h$	$p \times z = k$	$\bar{R}_{RP}$	$\bar{R}_{nRP}$	$\Delta \bar{R}$	$\bar{\bar{R}}_{RP}$	$\bar{\bar{R}}_{nRP}$
$30 \times 50 = 1500$	$0.3 \times 0.10 = 0.030$	6.1	6.1	0.1	0.2	0.4
	$0.4 \times 0.10 = 0.040$	5.5	5.2	0.3	0.3	0.2
	$0.5 \times 0.10 = 0.050$	5.0	4.7	0.4	0.1	0.2
	$1 \times 1 = 1$	4.0	2.7	1.3	0	0
$30 \times 100 = 3000$	$0.4 \times 0.05 = 0.020$	6.7	6.0	0.7	0.3	0.3
	$0.5 \times 0.05 = 0.025$	6.0	5.4	0.6	0.3	0.2
	$0.3 \times 0.10 = 0.030$	5.1	4.7	0.3	0.2	0.3
	$0.4 \times 0.10 = 0.040$	4.6	4.2	0.4	0.1	0.2
	$0.5 \times 0.10 = 0.050$	4.3	3.9	0.4	0.1	0.1
	$1 \times 1 = 1$	3.6	3.2	0.4	0	0
$30 \times 150 = 4500$	$0.3 \times 0.05 = 0.015$	7.0	6.3	0.7	0.4	0.2
	$0.4 \times 0.05 = 0.020$	6.1	5.5	0.6	0.2	0.3
	$0.5 \times 0.05 = 0.025$	5.6	5.0	0.6	0.3	0.2
	$0.3 \times 0.10 = 0.030$	4.7	4.5	0.2	0.2	0.2
	$0.4 \times 0.10 = 0.040$	4.3	4.2	0.1	0.2	0.1
	$0.5 \times 0.10 = 0.050$	4.1	4.0	0.0	0.1	0.1
	$1 \times 1 = 1$	3.4	3.8	-0.3	0	0
$60 \times 50 = 3000$	$0.3 \times 0.05 = 0.015$	12.7	12.2	0.4	1.0	1.0
	$0.4 \times 0.05 = 0.020$	11.4	10.1	1.3	0.7	1.0
	$0.5 \times 0.05 = 0.025$	10.3	8.5	1.8	0.5	0.8
	$0.3 \times 0.10 = 0.030$	9.2	7.0	2.2	0.4	0.5
	$0.4 \times 0.10 = 0.040$	8.5	5.8	2.7	0.2	0.3
	$0.5 \times 0.10 = 0.050$	8.3	5.4	2.9	0.3	0.2
	$1 \times 1 = 1$	8.1	5.2	2.9	0	0
$60 \times 100 = 6000$	$0.2 \times 0.05 = 0.010$	11.7	10.1	1.6	0.9	1.1
	$0.3 \times 0.05 = 0.015$	9.6	8.1	1.4	0.5	0.7
	$0.4 \times 0.05 = 0.020$	9.0	6.8	2.2	0.3	0.5
	$0.5 \times 0.05 = 0.025$	8.6	6.2	2.4	0.3	0.2
	$0.3 \times 0.10 = 0.030$	8.1	5.6	2.5	0.2	0.2
	$0.4 \times 0.10 = 0.040$	7.9	5.1	2.7	0.2	0.1
	$0.5 \times 0.10 = 0.050$	7.7	5.0	2.8	0.2	0.1
	$1 \times 1 = 1$	7.5	4.6	2.9	0	0
$60 \times 150 = 9000$	$0.2 \times 0.05 = 0.010$	10.0	8.4	1.7	0.6	0.7
	$0.3 \times 0.05 = 0.015$	8.6	7.1	1.5	0.4	0.4
	$0.4 \times 0.05 = 0.020$	8.2	6.4	1.8	0.3	0.3
	$0.5 \times 0.05 = 0.025$	7.9	6.0	1.9	0.2	0.2
	$0.3 \times 0.10 = 0.030$	7.5	5.5	2.0	0.2	0.2
	$0.4 \times 0.10 = 0.040$	7.4	5.2	2.2	0.1	0.2
	$0.5 \times 0.10 = 0.050$	7.3	5.1	2.2	0.1	0.1
	$1 \times 1 = 1$	7.2	5.0	2.2	0	0
$90 \times 50 = 4500$	$0.2 \times 0.05 = 0.010$	18.3	16.5	1.8	1.9	1.8
	$0.3 \times 0.05 = 0.015$	14.5	12.1	2.5	0.9	1.5
	$0.4 \times 0.05 = 0.020$	12.9	9.8	3.1	0.5	0.7
	$0.5 \times 0.05 = 0.025$	12.0	8.3	3.7	0.5	0.7
	$0.3 \times 0.10 = 0.030$	11.2	7.4	3.8	0.5	0.4
	$0.4 \times 0.10 = 0.040$	10.7	6.8	3.9	0.2	0.2
	$0.5 \times 0.10 = 0.050$	10.6	6.5	4.0	0.2	0.2
	$1 \times 1 = 1$	10.7	6.8	4.0	0	0
$90 \times 100 = 9000$	$0.2 \times 0.05 = 0.010$	13.3	10.4	2.9	0.9	1.1
	$0.3 \times 0.05 = 0.015$	11.5	8.1	3.4	0.4	0.4
	$0.4 \times 0.05 = 0.020$	11.0	7.5	3.5	0.3	0.3
	$0.5 \times 0.05 = 0.025$	10.8	7.1	3.7	0.3	0.3
	$0.3 \times 0.10 = 0.030$	10.5	6.9	3.6	0.2	0.2
	$0.4 \times 0.10 = 0.040$	10.4	6.6	3.8	0.2	0.1
	$0.5 \times 0.10 = 0.050$	10.3	6.5	3.8	0.1	0.1
	$1 \times 1 = 1$	10.4	6.5	3.9	0	0

(continued).

Table 3. Continued.

$w \times h$	$p \times z = k$	$\bar{R}_{RP}$	$\bar{R}_{nRP}$	$\Delta \bar{R}$	$\bar{\bar{R}}_{RP}$	$\bar{\bar{R}}_{nRP}$
$90 \times 150 = 13,500$	$0.2 \times 0.05 = 0.010$	11.7	8.8	2.9	0.7	0.6
	$0.3 \times 0.05 = 0.015$	10.9	7.7	3.2	0.4	0.2
	$0.4 \times 0.05 = 0.020$	10.6	7.3	3.3	0.3	0.3
	$0.5 \times 0.05 = 0.025$	10.5	7.1	3.4	0.2	0.2
	$0.3 \times 0.10 = 0.030$	10.3	6.9	3.3	0.2	0.2
	$0.4 \times 0.10 = 0.040$	10.2	6.7	3.4	0.2	0.1
	$0.5 \times 0.10 = 0.050$	10.1	6.6	3.5	0.1	0.1
	$1 \times 1 = 1$	10.1	6.5	3.6	0	0

Observation 5.1: The standard deviation is low, that is, all $\bar{\bar{R}}_{RP}, \bar{\bar{R}}_{nRP} \in [0.1\%, 0.7\%]$, except few cases whose density of probe data is low (for example, $k = 0.01$ and 0.015). In particular, when the density of the probe data is increased to be $k = 0.04$ or 0.05 , $\bar{\bar{R}}_{RP}, \bar{\bar{R}}_{nRP} \in [0.1\%, 0.2\%]$. It means that the results are stable and little influenced by the randomness in sampling, in particular when the density of probe data is larger than $k = 0.04$ or 0.05 .

Observation 5.2: With the growth of k , both $\bar{R}_{RP}$ and $\bar{R}_{nRP}$ decrease. When $k = 0.05$, the MAPE have been very close to the results using complete trajectory data, in particular when the cells are large. For example, one can compare the underlined rows with the double-underlined rows in Table 3. It turns out that $k = 0.05$ may be dense enough to represent complete data in constructing the SSC diagram when using the naive method. In other words, simply increasing the density of probe data may not be enough to improve the accuracy of estimating travel time when using the naive method.

6. Sensitivity analysis of wave speed estimations

Although it is known that the wave speed is about $-10 \sim -20$ km/h around the world, there is still 10 km/h difference between the maximum and minimum values. The section is dedicated to analyzing the sensitivity that we take different wave speeds as the slope of the left and right edges of the nRP cells, because in practice we may not exactly know the wave speed. To this end, we take the maximum and minimum wave speed¹ (i.e. -20 and -10 km/h) as examples, and first calculate the MAPE between the estimated and ground truth travel times. Then, compare the MAPE difference between using nRP cells with proper wave speed and using the over- and under-estimated wave speeds. Denote by $\bar{R}_{nRP}^{-10}$ and $\bar{R}_{nRP}^{-20}$ the MAPE resulted from using nRP cell with the edge slopes of -10 and -20 km/h, respectively. Similarly, the bar represents the average of 20 replications that are made when we use the probe data. The MAPE differences between $\bar{R}_{nRP}$ and $\bar{R}_{nRP}^{-10}$, between $\bar{R}_{nRP}$ and $\bar{R}_{nRP}^{-20}$, and between $\bar{R}_{nRP}^{-10}$ and $\bar{R}_{nRP}^{-20}$, are computed as follows:

$$\Delta \bar{R}_{nRP}^{-10} = \bar{R}_{nRP} - \bar{R}_{nRP}^{-10}, \quad (4a)$$

$$\Delta \bar{R}_{nRP}^{-20} = \bar{R}_{nRP} - \bar{R}_{nRP}^{-20}, \quad (4b)$$

$$\Delta \bar{R}_{nRP}^{-10,-20} = \bar{R}_{nRP}^{-10} - \bar{R}_{nRP}^{-20}. \quad (4c)$$

Table 4 presents the results, and it can be seen from the table as follows:

Observation 6.1: It can be generally seen from Table 4 that $\bar{R}_{nRP}^{-10} \in [2.6\%, 17.4\%]$ and $\bar{R}_{nRP}^{-20} \in [2.9\%, 17.1\%]$; the values are marked in bold in Table 4. It means that the travel time estimation errors by using the SSC diagram is not large, even when the wave speed is largely mis-estimated.

Table 4. Sensitivity analysis of using different wave speeds.

$w \times h$	$p \times z = k$	$\bar{R}_{RP}$	$\bar{R}_{nRP}$	$\bar{R}_{nRP}^{-10}$	$\bar{R}_{nRP}^{-20}$	$\Delta \bar{R}_{nRP}^{-10}$	$\Delta \bar{R}_{nRP}^{-20}$	$\Delta \bar{R}_{RP}^{-10,-20}$
$30 \times 50 = 1500$	$0.3 \times 0.10 = 0.030$	6.1	6.1	6.7	5.7	0.7	-0.4	1.0
	$0.4 \times 0.10 = 0.040$	<u>5.5</u>	5.2	<u>5.7</u>	5.0	0.5	-0.2	0.7
	$0.5 \times 0.10 = 0.050$	5.0	4.7	4.9	4.4	0.3	-0.3	0.5
	$1 \times 1 = 1$	4.0	2.7	2.6	2.9	-0.0	0.2	-0.2
$30 \times 100 = 3000$	$0.4 \times 0.05 = 0.020$	<u>6.7</u>	6.0	<u>7.2</u>	5.7	1.2	-0.3	1.5
	$0.5 \times 0.05 = 0.025$	<u>6.0</u>	5.4	<u>6.2</u>	5.2	0.8	-0.2	1.0
	$0.3 \times 0.10 = 0.030$	<u>5.1</u>	4.7	<u>5.3</u>	4.4	0.6	-0.3	0.9
	$0.4 \times 0.10 = 0.040$	<u>4.6</u>	4.2	<u>4.7</u>	4.0	0.5	-0.2	0.8
	$0.5 \times 0.10 = 0.050$	<u>4.3</u>	3.9	<u>4.4</u>	3.7	0.5	-0.3	0.8
	$1 \times 1 = 1$	3.6	3.2	2.8	3.6	-0.4	0.4	-0.8
$30 \times 150 = 4500$	$0.3 \times 0.05 = 0.015$	<u>7.0</u>	6.3	<u>7.5</u>	5.9	<u>1.2</u>	-0.4	1.6
	$0.4 \times 0.05 = 0.020$	<u>6.1</u>	5.5	<u>6.7</u>	5.2	<u>1.3</u>	-0.3	1.6
	$0.5 \times 0.05 = 0.025$	<u>5.6</u>	5.0	<u>6.1</u>	4.8	<u>1.1</u>	-0.2	1.3
	$0.3 \times 0.10 = 0.030$	<u>4.7</u>	4.5	<u>5.5</u>	4.2	<u>1.0</u>	-0.3	1.3
	$0.4 \times 0.10 = 0.040$	<u>4.3</u>	4.2	<u>5.2</u>	3.9	<u>1.0</u>	-0.3	1.3
	$0.5 \times 0.10 = 0.050$	<u>4.1</u>	4.0	<u>5.1</u>	3.7	<u>1.1</u>	-0.3	1.4
	$1 \times 1 = 1$	3.4	3.8	3.2	4.5	-0.6	0.8	-1.3
$60 \times 50 = 3000$	$0.3 \times 0.05 = 0.015$	12.7	12.2	12.5	11.6	0.3	-0.6	0.9
	$0.4 \times 0.05 = 0.020$	11.4	10.1	10.4	9.7	0.3	-0.4	0.7
	$0.5 \times 0.05 = 0.025$	10.3	8.5	8.4	8.0	-0.1	-0.4	0.4
	$0.3 \times 0.10 = 0.030$	9.2	7.0	7.0	7.0	-0.0	-0.0	0.0
	$0.4 \times 0.10 = 0.040$	8.5	5.8	5.8	6.0	0.1	0.2	-0.1
	$0.5 \times 0.10 = 0.050$	8.3	5.4	5.4	5.6	-0.0	0.2	-0.2
	$1 \times 1 = 1$	8.1	5.2	4.8	4.3	-0.4	-0.9	0.5
$60 \times 100 = 6000$	$0.2 \times 0.05 = 0.010$	11.7	10.1	10.5	10.1	0.4	0.0	0.4
	$0.3 \times 0.05 = 0.015$	9.6	8.1	7.8	8.0	-0.3	-0.1	-0.2
	$0.4 \times 0.05 = 0.020$	9.0	6.8	7.1	6.9	0.3	0.0	0.2
	$0.5 \times 0.05 = 0.025$	8.6	6.2	6.4	6.3	0.3	0.1	0.1
	$0.3 \times 0.10 = 0.030$	8.1	5.6	5.8	5.7	0.2	0.1	0.0
	$0.4 \times 0.10 = 0.040$	7.9	5.1	5.4	5.4	0.2	0.3	-0.1
	$0.5 \times 0.10 = 0.050$	7.7	5.0	5.2	5.3	0.2	0.3	-0.1
	$1 \times 1 = 1$	7.5	4.6	4.9	4.7	0.3	0.2	0.1
$60 \times 150 = 9000$	$0.2 \times 0.05 = 0.010$	10.0	8.4	8.9	8.4	0.5	0.1	0.5
	$0.3 \times 0.05 = 0.015$	8.6	7.1	7.3	7.0	0.2	-0.1	0.3
	$0.4 \times 0.05 = 0.020$	8.2	6.4	6.8	6.3	0.4	-0.0	0.4
	$0.5 \times 0.05 = 0.025$	7.9	6.0	6.6	6.0	0.6	0.0	0.6
	$0.3 \times 0.10 = 0.030$	7.7	5.5	6.1	5.6	0.6	0.1	0.5
	$0.4 \times 0.10 = 0.040$	7.4	5.2	5.9	5.4	0.6	0.2	0.5
	$0.5 \times 0.10 = 0.050$	7.3	5.1	5.8	5.3	0.7	0.1	0.5
$90 \times 50 = 4500$	$1 \times 1 = 1$	7.2	5.0	5.0	5.5	-0.0	0.5	-0.5
	$0.2 \times 0.05 = 0.010$	18.3	16.5	17.4	17.1	0.9	0.6	0.3
	$0.3 \times 0.05 = 0.015$	14.5	12.1	13.3	12.9	1.2	0.8	0.4
	$0.4 \times 0.05 = 0.020$	12.9	9.8	10.2	10.5	0.4	0.7	-0.3
	$0.5 \times 0.05 = 0.025$	12.0	8.3	8.7	9.0	0.4	0.7	-0.3
	$0.3 \times 0.10 = 0.030$	11.2	7.4	7.6	8.3	0.2	0.9	-0.6
	$0.4 \times 0.10 = 0.040$	10.7	6.8	7.0	7.7	0.3	0.9	-0.7
	$0.5 \times 0.10 = 0.050$	10.6	6.5	6.9	7.6	0.3	1.0	-0.7
$90 \times 100 = 9000$	$1 \times 1 = 1$	10.7	6.8	7.2	6.3	0.4	-0.4	0.8
	$0.2 \times 0.05 = 0.010$	13.3	10.4	10.8	10.9	0.4	0.5	-0.1
	$0.3 \times 0.05 = 0.015$	11.5	8.1	8.6	9.0	0.5	0.8	-0.3
	$0.4 \times 0.05 = 0.020$	11.0	7.5	7.9	8.3	0.4	0.8	-0.4
	$0.5 \times 0.05 = 0.025$	10.8	7.1	7.5	7.9	0.4	0.8	-0.4
	$0.3 \times 0.10 = 0.030$	10.5	6.9	7.2	7.7	0.3	0.8	-0.5
	$0.4 \times 0.10 = 0.040$	10.4	6.6	7.0	7.5	0.4	0.9	-0.5
	$0.5 \times 0.10 = 0.050$	10.3	6.5	6.9	7.4	0.4	0.9	-0.5
$1 \times 1 = 1$		10.4	6.5	7.2	6.6	0.7	0.1	0.6

(continued).

Table 4. Continued.

$w \times h$	$p \times z = k$	$\bar{R}_{RP}$	$\bar{R}_{nRP}$	$\bar{R}_{nRP}^{-10}$	$\bar{R}_{nRP}^{-20}$	$\Delta \bar{R}_{nRP}^{-10}$	$\Delta \bar{R}_{nRP}^{-20}$	$\Delta \bar{R}_{RP}^{-10,-20}$
90 × 150 = 13,500	0.2 × 0.05 = 0.010	11.7	8.8	9.3	9.4	0.6	0.7	−0.1
	0.3 × 0.05 = 0.015	10.9	7.7	8.2	8.3	0.5	0.6	−0.1
	0.4 × 0.05 = 0.020	10.6	7.3	7.8	8.0	0.5	0.6	−0.1
	0.5 × 0.05 = 0.025	10.5	7.1	7.7	7.8	0.6	0.7	−0.1
	0.3 × 0.10 = 0.030	10.3	6.9	7.4	7.7	0.5	0.8	−0.2
	0.4 × 0.10 = 0.040	10.2	6.7	7.2	7.5	0.5	0.7	−0.2
	0.5 × 0.10 = 0.050	10.1	6.6	7.2	7.4	0.6	0.8	−0.2
	1 × 1 = 1	10.1	6.5	7.2	7.0	0.7	0.5	0.2

Observation 6.2: In spite of being mis-estimated, all $\bar{R}_{nRP}^{-10}$ and $\bar{R}_{nRP}^{-20}$ are still smaller than $\bar{R}_{RP}$, except that $\bar{R}_{nRP}^{-10}$ is usually larger than $\bar{R}_{RP}$ when $w = 30$ sec; the percentage is 39/52, and the opposite cases are marked using underlines in Table 4. The largest difference is even around 3% compared with the absolute value of approximately 10%. It means that even when the wave speed is largely mis-estimated, using nRP cells still outperforms using RP cells, if the cells are not quite narrow in width.

Observation 6.3: The cells with 30 sec in width and 150 m in height have larger $\Delta \bar{R}_{nRP}^{-10}$ values (more than 1%), compared with other cases; see the values with wavy lines in Table 4. It means that the ‘slender’ cells are more sensitive to the wave speed, that is, they are more easily influenced by the slope of its left and right edges, because the duration of the vehicles traveling in such cells is less.

Observation 6.4: Except the cells with 30 sec in width and 150 m, $\Delta \bar{R}_{nRP}^{-10,-20} \in (-1\%, 1\%)$, $\Delta \bar{R}_{nRP}^{-10} \in (-0.5\%, 1\%)$, and $\Delta \bar{R}_{nRP}^{-20} \in (-0.5\%, 1\%)$, approximately. Compared with the magnitude of $\bar{R}_{RP}$ and $\bar{R}_{nRP}$ (i.e. 5 ~ 15% when using probe data), the ranges are acceptable in general, turning out that the results are not very sensitive to the wave speed.

7. Conclusion

This paper empirically evaluates and compares the basic method of using RP and nRP cells to construct the widely used SSC diagram, provided complete trajectory and partial probe data. Compared with RP cells, nRP cells additionally consider the backward-moving wave speed, whose importance has been demonstrated by recent research. US-101 trajectory dataset provided by NGSIM is employed to conduct the empirical study. Travel times that are directly estimated from the constructed SSC diagram are treated as metrics and are used to evaluate the quality of the SSC diagram. RP and nRP cells with various sizes are taken into account, and the effectiveness of the methods when providing complete trajectory and partial probe data is carefully compared. Sensitivity is analyzed by taking different wave speeds as the slope of the left and right edges of the nRP cells. The important results and findings are listed as follows:

- The errors of estimating travel time by using the SSC diagram composed of various-size cells is not large. The evidence includes:
When providing complete trajectory data, the MAPE of using RP cells ranges from 1.3% to 11.4%, the MAPE of using nRP cells ranges from 1.1% to 8.5%, and their difference (i.e. $R_{RP} - R_{nRP}$) ranges from −1.4% to 7% (Observation 3.1).
When providing probe data with the density that cannot result in large percentages (larger than 20%) of empty cells, the MAPE of using RP cells ranges from 4.1% to 18.3%, the MAPE of using nRP cells ranges from 3.9% to 16.5%, and their difference (i.e. $R_{RP} - R_{nRP}$) ranges from 0.1% to 4% (Observation 4.2).
Even when the wave speed is largely mis-estimated, the MAPE of using nRP cells ranges from 2.6% to 17.4% (Observation 6.1).

- Using nRP cells usually results in smaller errors than using RP cells. The evidence includes:
When providing complete trajectory data, most MAPE of using nRP cells, that is, 77 out of total 90 cases, are smaller (Observation 3.2).
When providing probe data with the density that cannot result in large percentages (larger than 20%) of empty cells, all MAPE of using nRP cells are smaller than the corresponding MAPE of using RP cells (Observation 4.3).
Even when the wave speed is largely mis-estimated (i.e. -10 or -20 km/h instead of the proper wave speed of -16 km/h), most cases (39 out of total 52 cases) have smaller MAPE when using nRP cells, and the largest MAPE difference between using nRP and RP cells is even around 3% compared with the absolute value around 10% (Observation 6.2).
- The outperformance of using nRP cells is more significant, when the traffic is in congestion with speed lower than 30 km/h, and when the cells are relatively large such that the width is larger than 60 s. The evidence includes:
When providing complete trajectory data, the MAPE difference between using nRP and RP cells could reach around 5%, compared with $R_{RP} \in (10\%, 11\%)$ and $R_{nRP} \in (4\%, 6\%)$ (Observations 3.3 and 3.5). In contrast, the difference is quite smaller or even is negative when the traffic speed is high (Observation 3.4).
When providing probe data, the difference is even as large as $2\% \sim 4\%$ in the condition that the total percentages are about 10% (Observations 4.4).
- When using probe data the stochastic impact from randomly sampling is small, indicated by the standard deviation smaller than 1%. In particular, when the density of probe data is large, the standard deviation is about 0.1% or 0.2% (Observation 5.1).
- Notably, with the growth of the density of probe data, the MAPE of using RP and nRP cells decrease. When the density is close to 0.05, the estimation quality of travel time has been very close to that using complete trajectory data (Observation 5.2).
- It is shown that both methods using RP and nRP cells over-estimate the travel time when the traffic speed is high, and under-estimate the travel time when the traffic speed is low. It is true for both cases using complete trajectory and probe data (Figures 5 and 6).
- The estimation results are not sensitive to the wave speeds generally, that is, the MAPE difference is less than 1% between using well-estimated wave speed and using largely mis-estimated wave speed (Observation 6.4). In contrast, when the cells are slender (i.e. with small width and large height; e.g. the width is 30 s and the height is 150 m), the MAPE difference may be over 1% (Observation 6.3).

In summary, the results indicate that using nRP cells outperforms using RP cells, in particular when the traffic is in congestion or when the cells are relatively large. Thus, no matter when providing complete trajectories or probe data, we recommend using nRP cells to construct the SSC diagram in future studies and applications, which is as simple as the traditional way of using RP cells but more accurate particularly in terms of estimating travel time. Notably, the prevailing map-based service providers, such as Google Map, usually evaluate traffic states and estimate travel time by using RP cells to aggregate instantaneous probe data. According to the results of the paper, we make a suggestion to use nRP cells.

Note

1. Although -20 is smaller than -10 in values, we still say the wave speed of -20 km/h is larger than the wave speed of -10 km/h, because the wave with the speed of -20 km/h propagates faster than that with the speed of -10 km/h.

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